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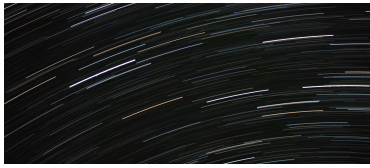
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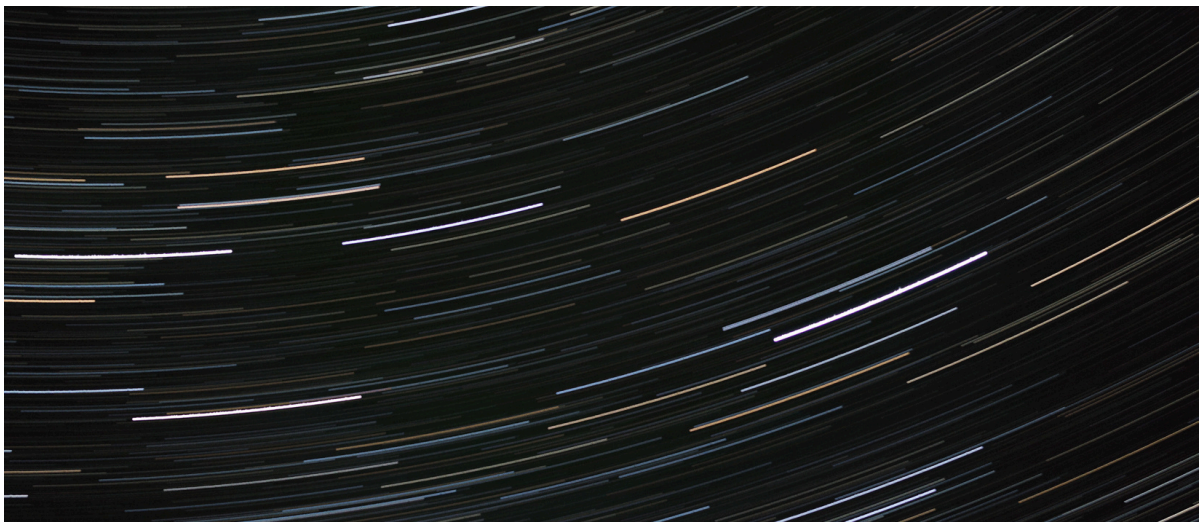
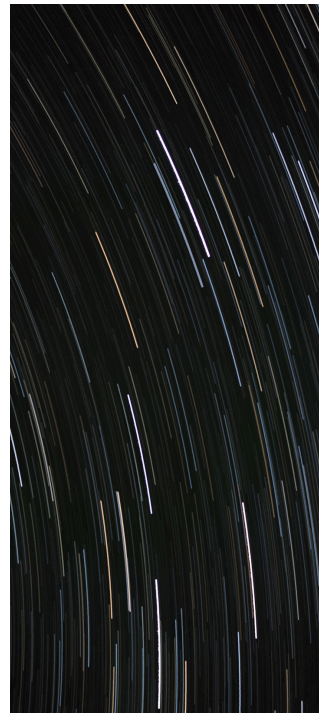
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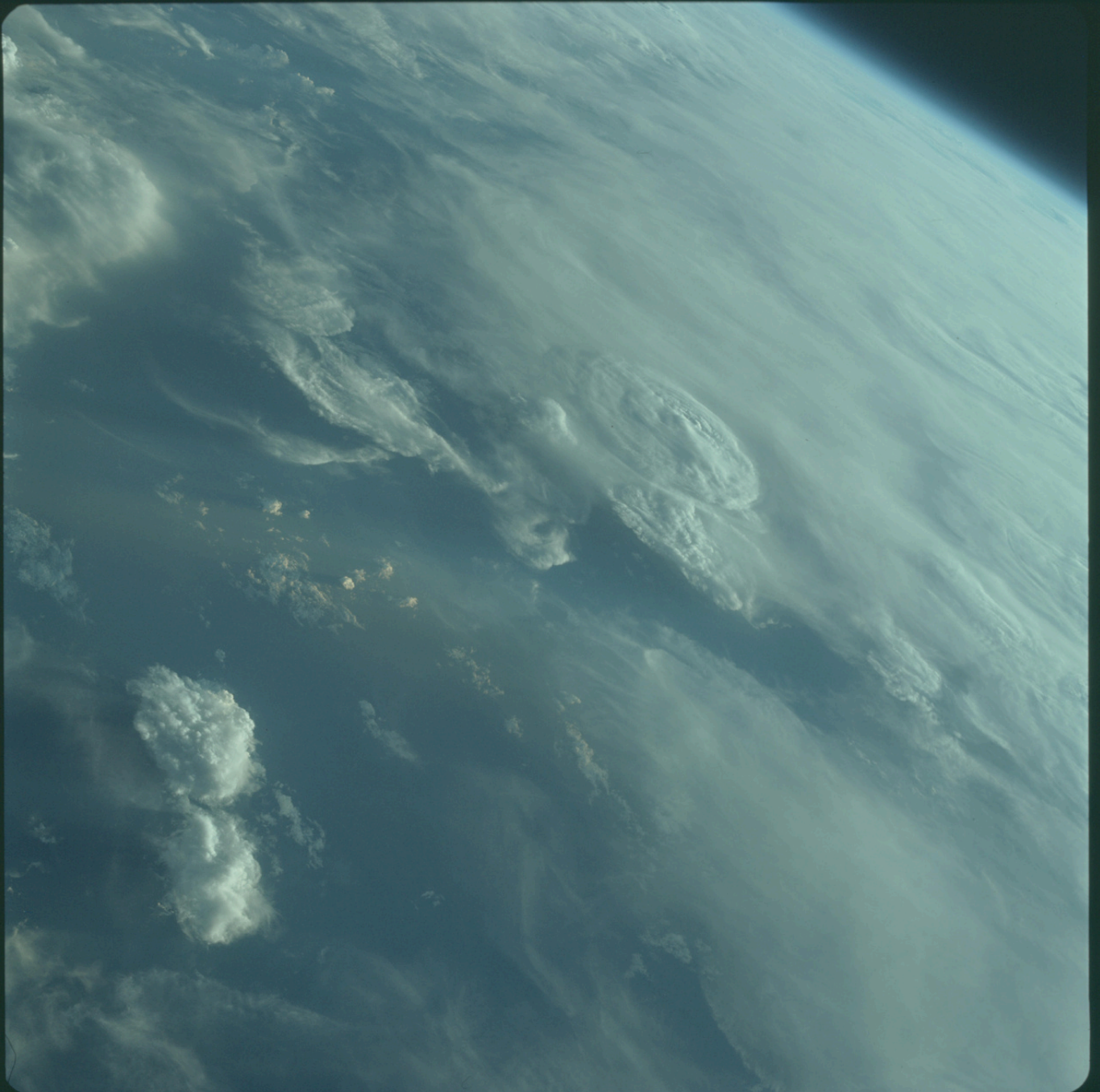
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Tune the orchestra

on the motion of the Earth - chapter IX

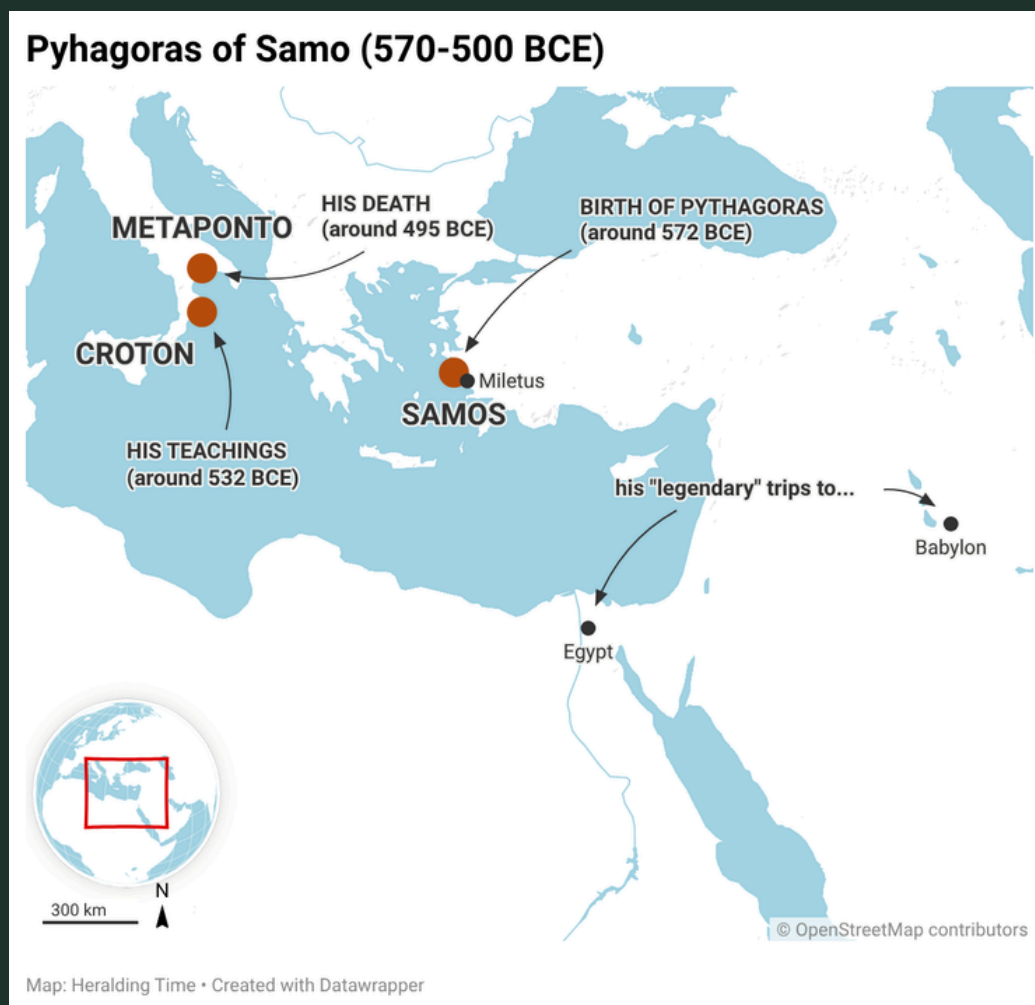
Arthur Koestler speaks of philosophy in Greece up to this moment as a musical ensemble in which everyone tries to tune their instrument. In the next episode we will talk about the one who managed to tune the orchestra.



Pythagoras: shaman, philosopher, myth

Historians are not certain that Pythagoras truly existed, because we have very little information about him — whether the life of which we have fragments belongs entirely to him, or whether several different figures live within his biography. Empedocles speaks of a wise man who lived through twenty human lives, although he may be referring to Parmenides. Apparently he may have lived between 570 and 500 BCE. It seems he was born in Samos in 572 BCE and died in Metapontum around 495 BCE.

According to Apollodorus, he was the son of Mnesarchus. He flourished around 532–531 BCE, a period during which he fled the tyranny of Polycrates. Around 530 BCE he moved to Croton, in southern Italy, where he founded a movement of both religious and philosophical character. At the end of his life he was banished from Croton and withdrew to Metapontum, where he died.



Some accounts suggest he traveled to Asia Minor, Egypt, and perhaps Babylon, but these are stories found in relation to other philosophers as well — and may be tales told merely to lend credence to the idea of a wisdom and knowledge originating from Asia.

Sources contemporary with Pythagoras reveal little about his intellectual activity. Xenophanes mentions metempsychosis, the philosophy of the transmigration of souls, though without explicitly naming Pythagoras. Heraclitus associates him with Polymatheia and praises him for his *historia*, that is, his inquiry, though the specific type of inquiry remains unclear.

None of these authors mentions mathematics, the theory of the harmony of the spheres, or other astronomical and cosmological theories that we today attribute to the Pythagorean school. From direct accounts, only traces of moral teachings survive.

No written works by Pythagoras himself remain — a circumstance that is not due to the secretive nature of his teachings, but simply to the fact that he wrote nothing. Before the era of Philolaus there was no written evidence of his doctrine, according to Plutarch.

Surprisingly, Plato and Aristotle mention him directly only three times in their writings, referring in any case to the Pythagoreans in general.

When we speak of the Pythagoreans, therefore, we will try to attribute each theory to its specific author, though this is not always easy. The person for whom we will do so most frequently is Philolaus, and we will discuss him at length. For the rest, we will speak of Pythagoreans in general.

Pythagoras and the Spherical Earth

For the Pythagoreans, the Earth is spherical, and they justified this claim in three ways: first, by observing the shadow the Earth casts on the Moon during lunar eclipses, which is always circular; second, in their celestial system the heavens separate different paradises, which by nature are all spherical — as is the cosmos itself; and finally, they had purely mathematical reasons. Diogenes Laërtius also attests that Pythagoras taught that the world was a sphere inhabited even at the antipodes, and that he was the first to call the heavens κόσμος (*kósmos*).

Science and religion are two inseparable factors in life, say the followers of Pythagoras. By contemplating the order of the universe, man can purify himself. They transform the physical ecstasy of Orpheus into the purely immaterial ecstasy of intellectual discovery. The word θεωρία (*theoria*) derives from *theoris*, the spectator, and *thea*, the spectacle: the Pythagoreans were contemplatives in the deepest sense. The man of science is above all one who observes.

The teachings of Pythagoras generated two distinct groups: the ακουσματικοί (akousmatikoi), those who listen, and the μαθηματικοί (mathematikoi), those who study. The former were religious, "conservative," and dogmatic in orientation, while the latter were scientific, "progressive," and intellectually active. The mathematikoi maintained that numbers were superior to matter, that mathematical definitions were superior to physical descriptions, and that rigorous demonstrations surpassed the "probable stories" of the Ionians.

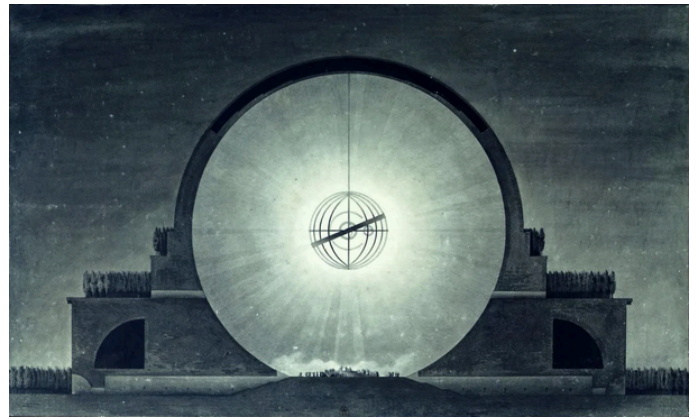
For the Pythagoreans, number was the principle, the ἀρχή (arche). Among the various numbers, the importance of the number ten derived from the fact that the principal musical intervals could be expressed by simple ratios among the numbers 1, 2, 3, and 4. The sum of these four numbers is precisely 10, a number that seems to encompass, as Aristotle and Aetius report, the entire nature of numbers.

Aristotle, in the *Physics* and the *Metaphysics*, describes how the Pythagoreans constructed their system: from numbers originate points, from lines the plane, from planes solids, and from solids sensible bodies. The latter are composed of fire, water, earth, and air — the elements that make up the κόσμος.

On the subject of κόσμος. Two central concepts govern Pythagorean thought: κόσμος (order, symbol of regulation and clarity) and Ἀρμονία (harmonia) (to put together, to connect). The κόσμος depends on harmonia through numbers, which are the principle of the world. Harmony is not merely an abstract concept: it is the first step in Western philosophy in the reduction from the qualitative to the quantitative — a reduction that occurs for the first time in music.

Plato reports that for the Pythagoreans, astronomy and harmony were sister sciences. μαθηματικά (mathematiká) encompassed geometry, arithmetic, number theory, astronomy, and harmonics — but also numerology.

The Harmony of the Spheres



The theory of the harmony of the spheres has no precise originator: we know that it lies at the heart of Pythagorean doctrine, and that it arises from the concordance between the numerical structure of the seven strings of the heptachord and the system of the seven planets. According to Pythagorean doctrine, the Earth is a sphere at the center of the universe. The Sun, the Moon, and the planets revolve around it in concentric spheres, each fixed to its own sphere. According to Theon of Smyrna, it is precisely the Pythagoreans who first define the concept of the celestial sphere.

The movements of these celestial bodies create music that spreads through the air: each planet sounds at a different frequency, determined by the length of its orbit. These orbits are governed by universal harmony, such that the totality of the music produced is concordant and in harmony. According to tradition, only Pythagoras himself was capable of hearing this music.

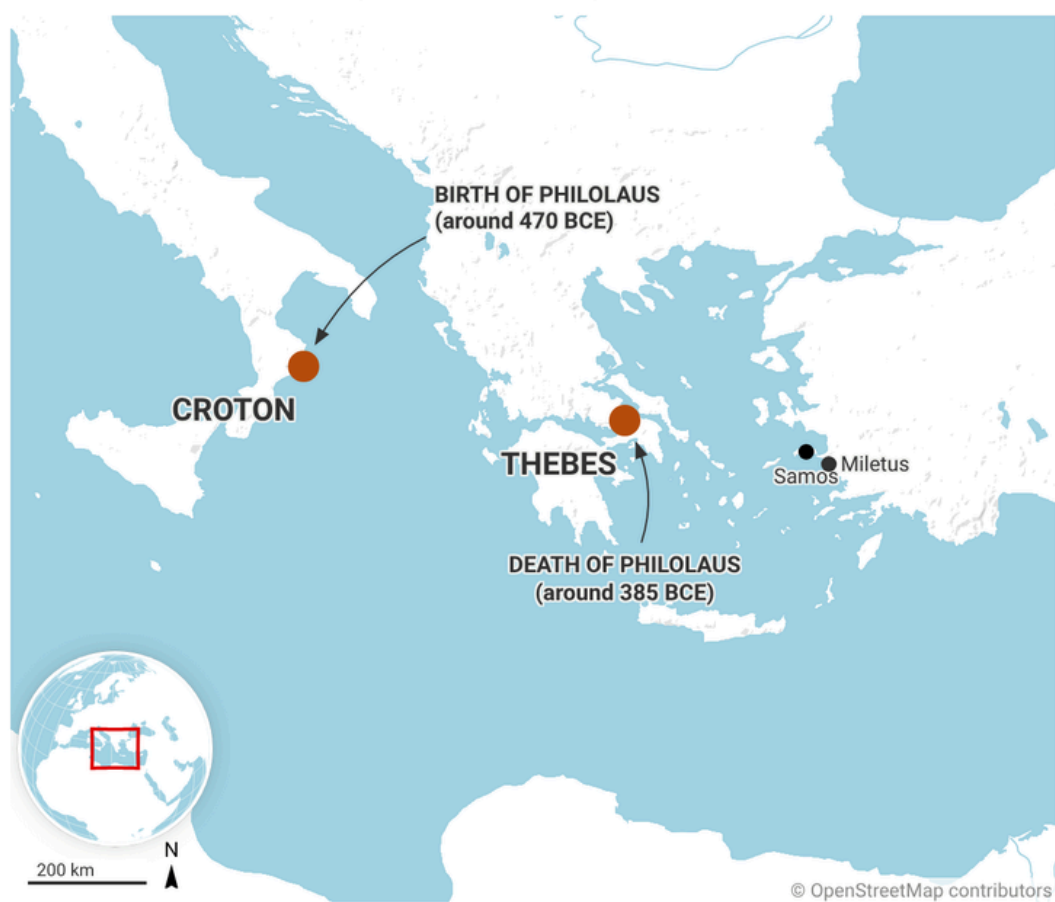
The orbits of the celestial bodies form an enormous lyre with curved strings. It is difficult to determine whether this was a genuine scientific theory or a poetic artifact. Aristotle considered it a fantasy, and yet Kepler was fascinated by it two thousand years later.

Our information about Pythagorean physics and cosmology depends largely on two sources: Aristotle and later Hellenistic sources such as Diogenes. Both speak of Philolaus, the philosopher who more than any other deserves credit for having devised the astronomical system accepted by the Pythagoreans. Aristotle observes that according to the Pythagoreans, the Earth was not considered sufficiently noble to occupy the center of the universe — a position that, ironically, it shares with the heliocentric system of later ages.

Philolaus: the creator of the Pythagorean astronomical system

Philolaus, born probably around 470 BCE and died around 385 BCE, is the most prominent follower of Pythagoras, second only to the founder himself in the Pythagorean tradition. A contemporary of Socrates, Plato reports that he died shortly before Plato's own birth. He taught in Thebes toward the end of the fifth century BCE, and a work entitled *On Nature* (Περὶ Φύσεως — *Perí Phýseos*) is attributed to him, perhaps also known as *On the World* (Περὶ τοῦ Κόσμου — *Perí tou Kosmou*), according to Diogenes Laërtius.

Philolaus of Croton (470-385 BCE)



Map: *Heralding Time* • Created with Datawrapper

Archytas of Tarentum cites Philolaus in his writings, but Philolaus cites no predecessor. This suggests that Philolaus was the first Pythagorean to provide a systematic account of his master's doctrine. According to Walter Burkert, many theories that Aristotle attributes to Pythagoras are in fact the work of Philolaus and Archytas.

G.E.R. Lloyd further argues that there is insufficient evidence that Pythagoras himself was a mathematician: only Archytas and Philolaus are clearly identifiable as both Pythagoreans and mathematicians within the tradition. With Philolaus, the idea of alternative cosmic geometries to the geocentric model began to spread, and it became common to distinguish between the daily rotation and the annual motion of the planets.

Philolaus initially states that everything is composed of the limited (peras) and the unlimited (apeiron). These are not simple abstract concepts but active powers: the capacity to limit and the power to be limited. Nature emerges from limited things (perainonta) and unlimited things (apeira), whose union generates harmony (harmonia). These three dimensions unite to create that which fits together (ἄρμόζον — armoxon — that which connects).

According to Irini-Fotini Vietanioti in her translation of Philolaus, "nature in the order of the world was brought together from the unlimited and the limited, both the inner order of the world and all things in it." However, it remains uncertain what exactly these limitings are: numbers, powers, or something else?

The universe itself, being ordered and bounded, is finite. It must have a physical boundary and a defined form. Void separates things within the cosmos, and beyond the sphere of the fixed stars there exists infinite space — infinite air — so that the cosmos may "breathe," recalling Anaximenes.



The Astronomical System

The Pythagoreans were the first to remove the Earth from the center of the universe. They did not do so for religious or metaphysical reasons, but for astronomical ones. This intellectual achievement, which anticipates by nearly two thousand years the Renaissance theories of Copernicus and Kepler, begins here. And the creator of this celestial system is Philolaus. He asserts that it is the Earth that revolves around a central point, not the sky that revolves around the Earth. The rotation of the Earth is daily, and this gives rise to the apparent daily rotation of the heavens — an optical illusion caused by the Earth's own motion around an external point. In this scheme, however, the Earth does not rotate on its own axis: it orbits around a central fire.

Daniel W. Graham coined the term "hestiocentrism" to describe this system, distinguishing it from both geocentrism and heliocentrism. The central fire does not illuminate: it is simply the focal point around which all celestial bodies orbit. Carl Huffman calls it "speculative astronomy," while Walter Burkert observes that "there is nothing mathematical about Philolaic astronomy."

The universe of Philolaus is spherical and finite in space, possessing defined dimensions. Beyond this finite sphere there exists infinite void, and the sphere has the central fire within it. Around this fire, in order of increasing distance, orbit the Counter-Earth (antichthon) — probably in an inner orbit between the Earth and the central fire — the Earth, the Moon, Mercury, Venus, the Sun, Mars, Jupiter, and Saturn, and finally the sphere of the fixed stars, at the boundary of the universe. Beyond the sphere of the fixed stars there exists an additional outer fire, and beyond that still, the infinite space mentioned earlier.

What is the Counter-Earth? We will address this shortly.

According to Plutarch, the original ordering of the planets according to the Pythagoreans was: Earth, Moon, Mercury, Venus, Sun. In contrast, Anaxagoras, Plato, and many other philosophers proposed the following ordering: Moon, Sun, Venus, Mercury, Mars, Jupiter, Saturn.

The Earth moves eastward, meaning that, viewed from the north pole, it moves counterclockwise. It completes one revolution around the central fire each day — in astronomical terminology, a "revolution year" is equivalent to a "rotation day." This daily motion of the Earth creates day and night, owing to the relative positions of the Sun and the Earth during their respective motions.

The Earth moves in the plane of the equator; the Sun and Moon, however, move in the plane of the zodiac, an orbit inclined with respect to that of the Earth. This inclination explains the succession of the seasons: the Sun traces an oblique circle as it orbits around the central fire.

The celestial bodies do not all orbit at the same speed: the Moon completes an orbit moving at 12° per day, orbiting approximately every two months; the Sun moves at about 1° per day, completing an orbit in one year; the planets move more slowly, with decreasing speed as distance from the fire increases; the sphere of the fixed stars moves very slowly. The Earth "overtakes" both the Sun and the Moon once a day, moving more quickly than both.

It is evident that in the Philolaic system the speed of the planets decreases as distance from the central fire increases — suggesting an intuitive grasp of a relationship between distance and orbital speed. Consequently, the Sun and Moon rise in the east and set in the west not because the sky rotates, but because the Earth "overtakes" them in its faster motion.



The Counter-Earth

The Counter-Earth represents one of the most enigmatic aspects of the Philolaic system. Aristotle suggests it was postulated for numerological reasons: with the Counter-Earth, the total number of celestial bodies reaches the perfect number of ten (together with the Sun, the Moon, the five planets, the Earth, and the sphere of the fixed stars). However, the Counter-Earth also carries significant astronomical implications.

One consequence of postulating an additional celestial body relatively close to the Earth is that it explains the apparent discrepancy between the number of lunar and solar eclipses. If the Moon can pass through both the shadow of the Earth and the shadow of the Counter-Earth, the number of lunar eclipses would be greater than that of solar eclipses — something that matches observations. Furthermore, the presence of the Counter-Earth could explain why lunar eclipses can occur at dawn or dusk, as studied by Anaxagoras.

A geographical consequence is also that lands beyond the Pillars of Hercules lie in eternal shadow, since the Counter-Earth hides the central fire from that region. Herodotus in fact reports the existence of peoples in the north who sleep for six months a year, as they dwell in perpetual night for half the year.

As for the Counter-Earth's position within the Philolaic system, sources diverge: some place it in the same orbit as the Earth, but 180° away; others on an inner orbit, hence closer to the central fire. A small distance between the Earth and the Counter-Earth would have mitigated one of the system's problems: the daily motion of the Earth would have caused the Earth–Sun and Earth–Moon distances to vary greatly, in turn greatly varying the apparent diameters of the Sun and Moon in the sky.

Light and the Celestial Bodies

In the Philolaic system, the Sun is not an original source of light. It is spherical and transparent like glass, and receives its light from the central fire. Diogenes Laërtius attests that the Moon has its own light by reflection from that of the Sun — a concept taken directly from Parmenides and developed further by Anaxagoras.

The central fire is not, however, the only source of light in the observable universe. There is also the "fire above," beyond the perceptible universe — that which lies beyond the sphere of the fixed stars. This upper fire may have a conical shape. Millennia later, for Giovanni Virginio Schiaparelli this configuration resembles the zodiacal light; for Eduard Zeller, it resembles the shape of the Milky Way as seen from the outside. The Milky Way itself is conceived as the shadow the Sun casts when reached by the light of the central fire.

Stobaeus attests that some Pythagoreans believed the visible markings on the Moon were the reflection of the Earth's seas, and others say that for Philolaus the Moon is part of the Earth.

Philolaus accepts the theories of heliophotism (that the reflection of sunlight causes the phases of the Moon) and antiphraxis (that eclipses are caused by the blocking of sunlight by the Moon or the Earth). Lunar eclipses occur both when the Moon passes through the shadow of the Earth and when it passes through the shadow of the Counter-Earth.



The Problems

The Philolaic system presents two categories of problems. First, the Counter-Earth appears to serve no explanatory function beyond numerology and the explanation of eclipses, and there is no way to verify its existence empirically; second, how is the motion of the universe to be explained?

While for the first issue one might raise the objection that the empirical verification of astronomical theories was almost never the guiding compass of Greek astronomers, the second creates problems even from a purely philosophical standpoint: is the motion of the universe single, or are there additional motions proper to individual planets? Does the sphere of the fixed stars move everything except the Earth, which possesses its own proper motion? And what speed does the sphere of the fixed stars possess?

If the harmony of the spheres exists, these questions are not merely technical in character, but take on a moral dimension, since every body participates in the harmony of the cosmos in which we are immersed.

What the system of Philolaus does explain — such as day and night, the daily rotation of the heavenly bodies, the motion of the Moon, the planets, and the Sun through the zodiac, and the succession of the seasons — makes it the system that provides more explanatory power than was generally required of cosmological models, at least for its era.

Other contributions concern the manner of thinking about the sky and the celestial bodies. The Philolaic system defines the concept of the center of revolution of an orbit as a geometric point distinct from the physical center of a body; it clearly distinguishes between planets and other celestial objects, and accustoms astronomers and philosophers to the idea of a spherical Earth.

Yet, as Aristotle observes in the *Physics* and the *Metaphysics*, the Pythagoreans "filled the gaps in their theory to make their system coherent," adapting phenomena to their philosophical and mystical predilections rather than the other way around. They were certainly not the only ones to do so, but they may have been the first.

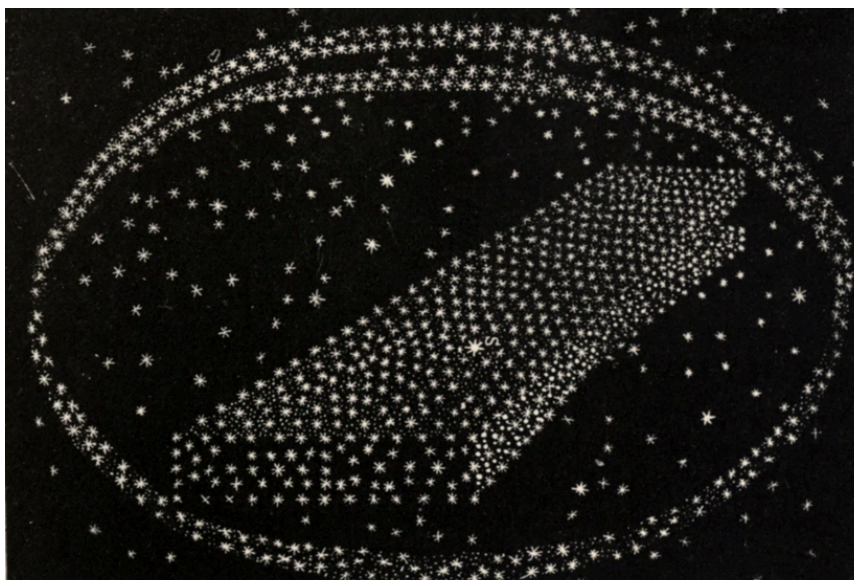
Modifications, Variations, Impossible Attributions

The theory of Philolaus underwent subsequent modifications. The number of celestial bodies increased beyond ten, the Counter-Earth was eliminated, and the central fire was located at the center of the Earth itself — a clever way of avoiding the need to locate it in the sky.

About six hundred years after the era of Philolaus, when his figure had already become legendary, a revised version of the astronomical system was attributed to the Pythagoreans — one in which the Earth is once again at the center of the Universe.

We have said that the concept of the harmony of the spheres is owed to the Pythagoreans: this, together with the discovery that Hesperus, the evening star, and Phosphorus, the morning star, are the same star (Venus), may also be discoveries attributable to Parmenides. The attribution of the Pythagorean theorem is analogous: although it bears the name of Pythagoras, the theorem and the corollary concerning the incommensurability of the diagonal of the square are not certainly his original discoveries.

The idea of the central fire was abandoned over the centuries, probably when the Greeks encountered peoples from other parts of the world — from the Ganges, from the mouth of the river Tagus, from Thule (an imaginary island in the North Atlantic, probably Iceland), from Taprobane (the modern island of Sri Lanka) — at whose coordinates there was no trace of the Counter-Earth. Some Pythagoreans then redefined the Counter-Earth, associating it with the Moon.



The cosmological construction of Philolaus represents a philosophical act of profound importance: it is the attempt to subordinate the facts of natural phenomena to coherent mathematical principles. If this sometimes entails forcibly adapting phenomena to theory rather than the reverse, the pursuit of a universal harmony in nature remains significant — and is something that still nourishes those who live by science.

Pythagoras Inaccessible

W.K.C. Guthrie notes that "the real Pythagoras is today more or less totally inaccessible." G.E.R. Lloyd emphasizes that there is insufficient evidence that he was a mathematician, a shaman, or a purely charismatic figure. What we know of Pythagoras's teachings comes to us principally through Plato and Aristotle, who however do not directly attribute the technical details of the astronomical theories to Pythagoras himself, citing "the Pythagoreans" in general.

A fundamental principle of Pythagorean cosmology emerges from Aristotle, who cites Alcmaeon of Croton: all divine things must be in motion. Therefore, even the sphere of the fixed stars, though it moves very slowly, must possess its own proper motion.

Pythagoreanism constitutes an independent movement, born in southern Italy, contemporary with the Ionian movement on the western coast of Asia Minor. It lasted approximately two hundred years and came to an end in the fourth century BCE after numerous persecutions. All discoveries were attributed to the founder — a practice that has obscured the true history of the movement and made it nearly impossible to associate any single theory with its actual originator.

Mathematics is for the Pythagoreans sacred and enriching, and represents the method of access to ultimate truth. Two visions of the world thus stand in opposition: the Ionians against the Pythagoreans, the materialists against the mathematicians, and further: things in relation against the relations themselves, atoms against patterns, substance against form, and structure against function.

The history of Pythagorean cosmology, from Pythagoras to Philolaus, represented the first systematic Western attempt to provide a quantitative and mathematical interpretation of celestial phenomena. Although the heliocentric system of Philolaus did not represent as great an advance toward a more accurate understanding of the universe as the geocentric model of the same period, it constituted a crucial step in the liberation of the human mind from the limitations of immediate sensory experience. In this sense, Pythagoras and Philolaus were the true precursors of the scientific revolution, anticipating by millennia the principle that underlying reality is mathematical, and that humanity can investigate it through mathematics.

The real Pythagoras is inaccessible, but his movement generated a universe ordered according to harmony and number — a cosmos in which geometry and motion describe the ultimate nature of things.

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